



School of Engineering

Course : Diploma in Electronic & Computer Engineering (EGDF20)

Module : EGE322 – IoT System Project

Laboratory 7: Connecting the ESP32 to the Internet Using ThingSpeak

1. Objectives

In this session, you will learn how to setup and configure ThingSpeak to display data from the ESP32 microcontroller board.

At the end of this laboratory session, you will be able to:

- set up IoT platform
- on and off the leds using a momentary switch.
- connect the ESP32 microcontroller to the OLED display.
- connect the ESP32 microcontroller to the keypad.

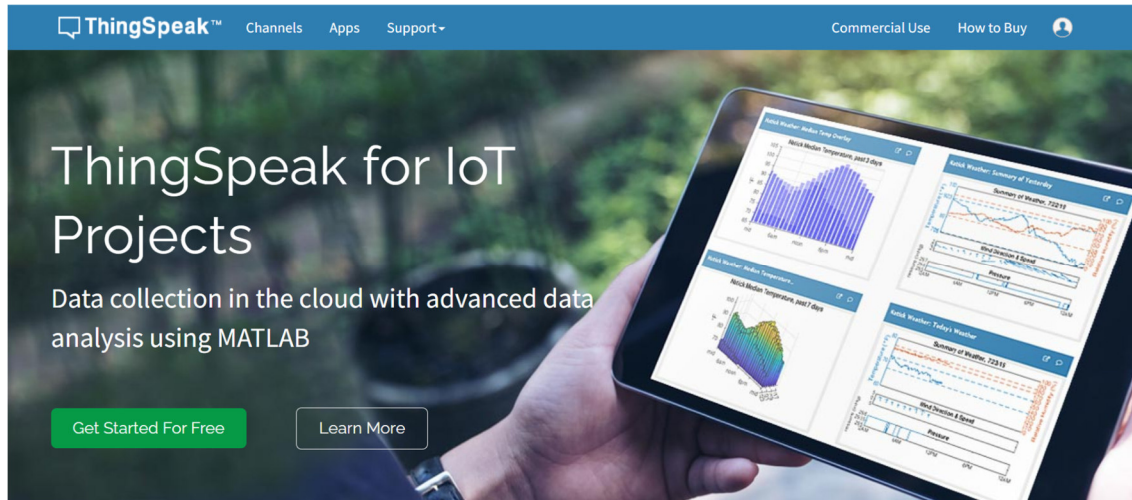
2. Material

- ✚ Window 10 x64 computer with USB port and Internet connectivity x1
- ✚ ESP32 NodeMCU microcontroller board x 1
- ✚ USB 2.0 data cable with mini-USB connectivity. x 1
- ✚ DHT22 Temperature and Humidity Sensor x1
- ✚ 24 x 24mm OLED display x 1

3. ThingSpeak

What is ThingSpeak?

ThingSpeak is an IoT analytics platform service that allows one to collect, visualize and analyze live data streams in the cloud. It provides instant visualisations of data from your computing devices or sensors.



4. Procedures to send data to the Internet using ThingSpeak

4.1 Registration and Setup

To begin using ThingSpeak, you will need to register an account with Thingspeak and setup the channel to display your data.

- Navigate to ThingSpeak (thingspeak.com) and sign up a new account or login to your existing account.
- Click on Sign Up at the top menu bar to sign up a new account:

Sign up for ThingSpeak

It is free to sign up for ThingSpeak. Free accounts offer a fully functional experience on ThingSpeak with limits on certain functionality. Commercial users may sign up for a time-limited free evaluation. To send data faster to ThingSpeak or to send more data, consider our [paid license options](#) for commercial, academic, home and student usage. To start using ThingSpeak you must create a new MathWorks account, or, click cancel and log in using an existing MathWorks account.

Create MathWorks Account

Email Address

To access your organization's MATLAB license, use your school or work email.

Location

First Name

Last Name

Diagram: SMART CONNECTED DEVICES → DATA AGGREGATION AND ANALYTICS (ThingSpeak) → MATLAB (ALGORITHM DEVELOPMENT SENSOR ANALYTICS)

Or Sign in to MathWork if you already have an account:

Sign In - ThingSpeak IoT

Sign in to your MathWorks Account

Email Address:

Password:

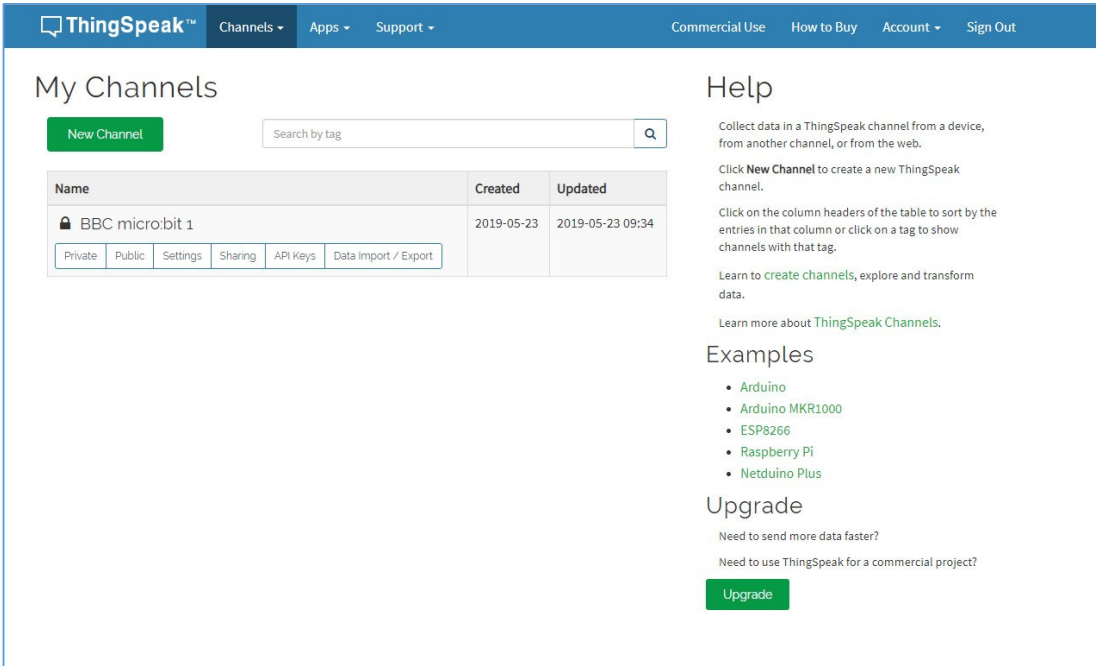
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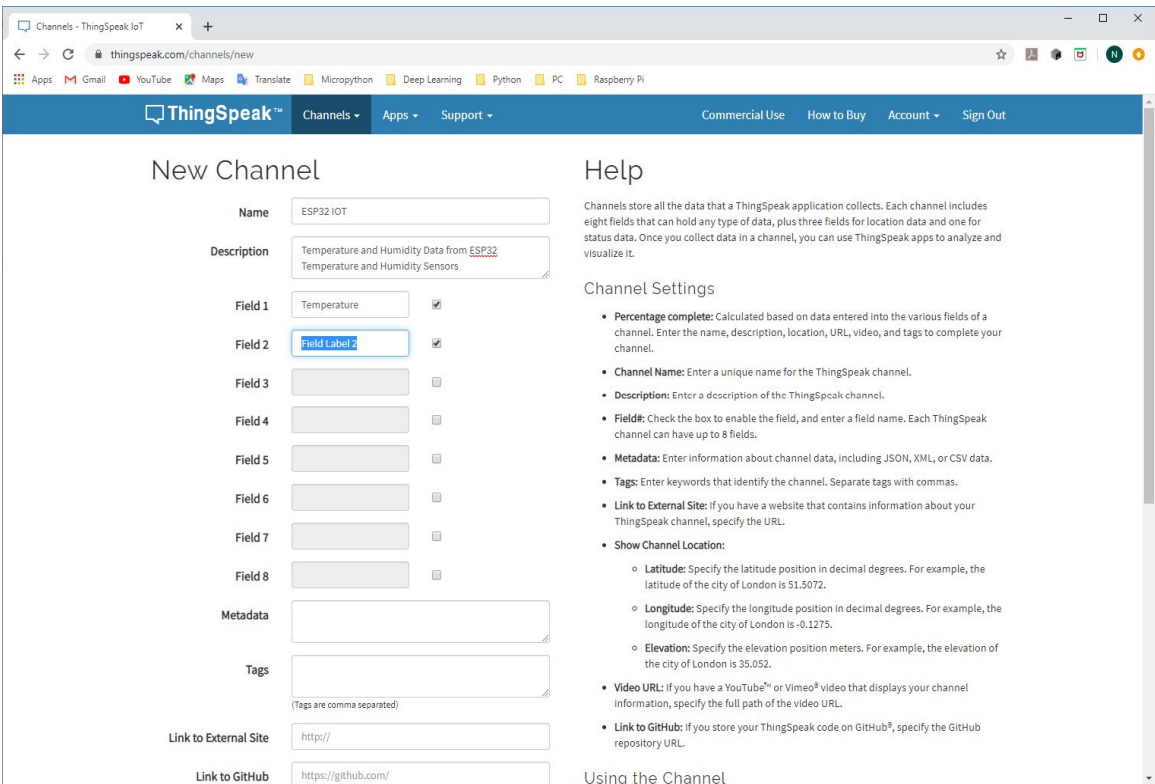
c) **Set Up a New Channel**

After you have signed in, you will be brought to a new page:

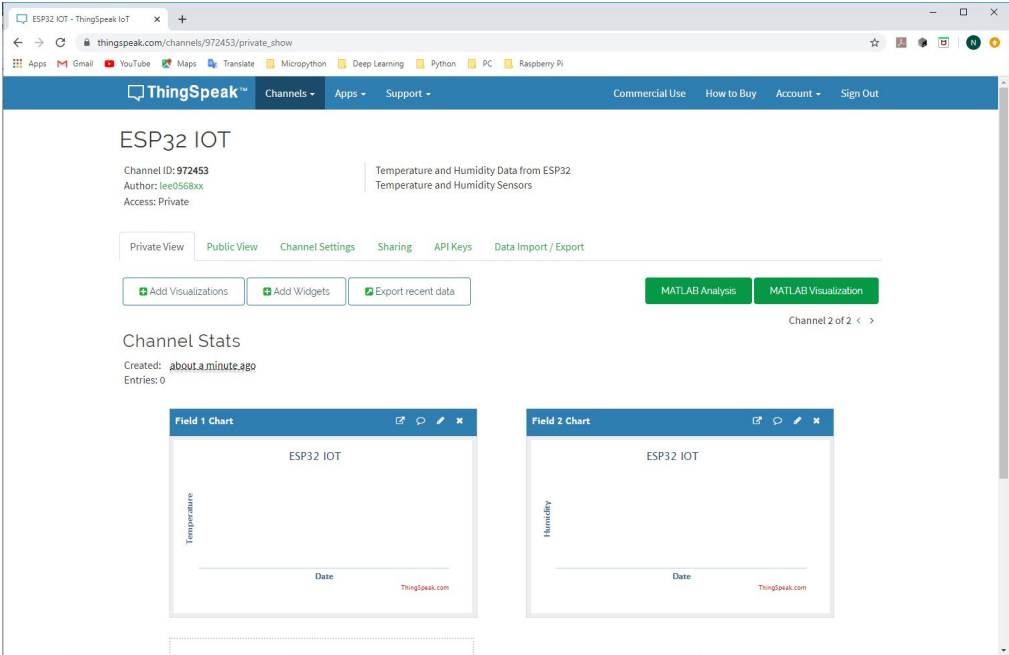


d) If you click on the New Channel button, you will be brought to a new page:

e) Enter the name of your new channel, description and 'tick' to enter the name of your data you want to submit to ThingSpeak in Field 1, 2, etc. You can submit up to 8 different data.

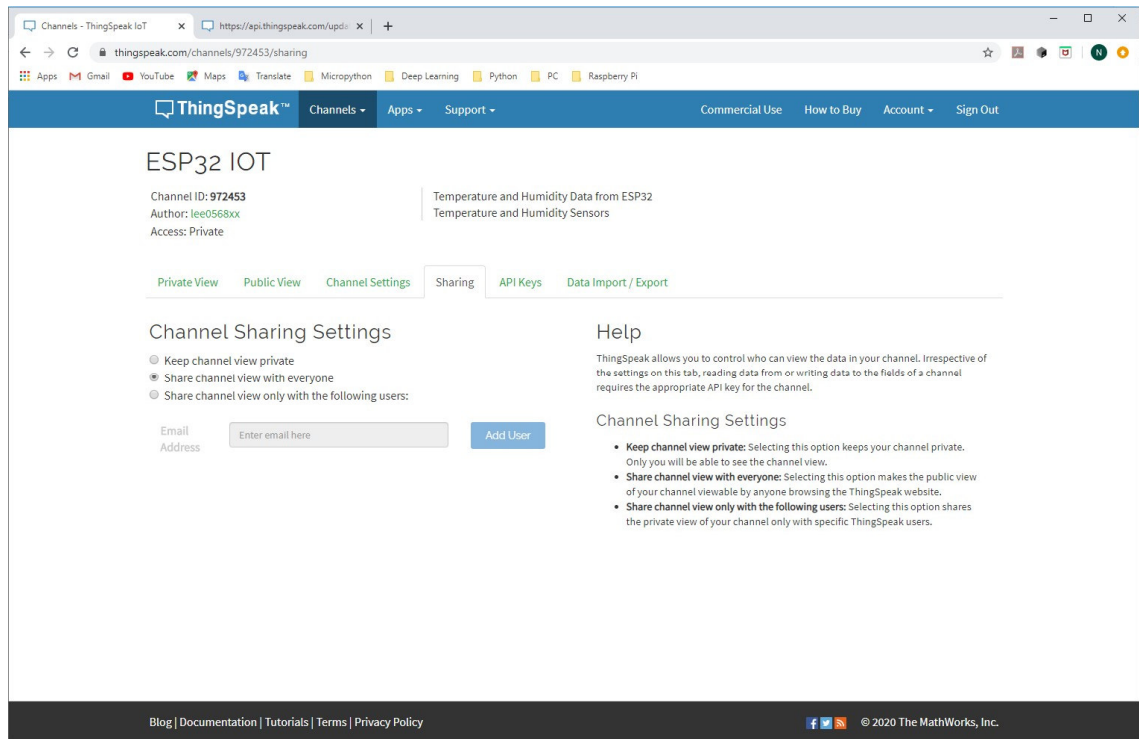


f) Click on 'Save Channel' at the bottom of the page.

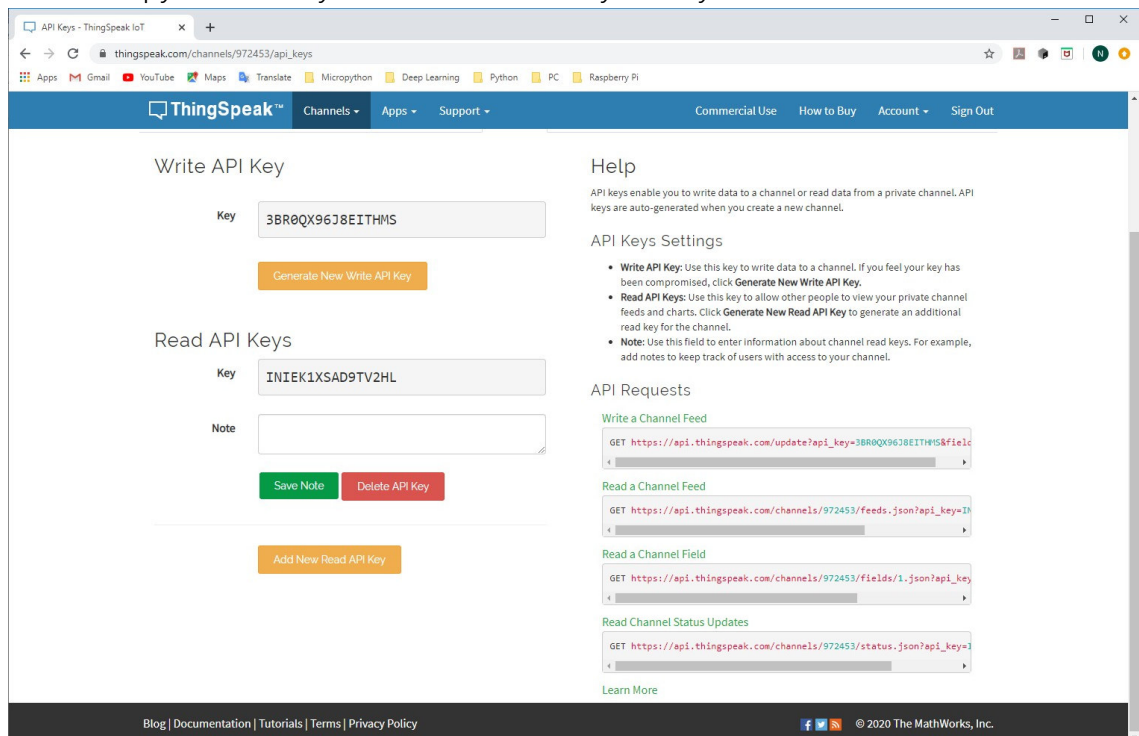


Private View	This tab displays information about your channel that only you can see.
Public View	If you choose to make your channel publicly available, use this tab to display selected fields and channel visualizations.
Channel Settings	This tab shows all the channel options you set at creation. You can edit, clear, or delete the channel from this tab.
Sharing	This tab shows channel sharing options. You can set a channel as private, shared with everyone (public), or shared with specific users.
API Keys	This tab displays your channel API keys. Use the keys to read from and write to your channel.
Data Import/ Export	This tab enables you to import and export channel data.

g) Click on Public View and enable sharing.



h) To submit data to the ThingSpeak cloud and assign them to a specific field, we have to copy our API Key link. Click on 'API Keys' and you will see:



- i) You will need your Channel ID and Write API Key in order to send your data from your ESP32 to ThingSpeak. You must NOT review your Write API Key to anyone. Generate a New Write API key if you suspect the security has been compromised.
- j) View Your Published Data in ThingSpeak

Data can be published on ThingSpeak using Restful API or MQTT. Thingspeak has a MQTT broker at the URL mqtt.thingspeak.com. It supports both MQTT publish and subscribe mode. Your Write-API key is required to publish data to ThingSpeak.



4.2 Uploading ESP32 Sensor Data using RESTful API

Before you can display your data in Thingspeak in previous steps, you will need to setup your ESP32 microcontroller with the correct peripheral devices and program the micropython script to send data to ThingSpeak using Restful API.

There are different types of Rest API. In this exercise, we will use rest API to send data to the internet IoT platform ThingSpeak.

- a) To begin, you can enter the micropython code from our previous exercise to connect the DHT22 and OLED display. You can connect the ESP32 to the DHT22 and OLED display using the Pin connections shown in the script below.

<u>DHT22</u>	<u>ESP32</u>
Pin 1	3.3V
Pin 2	GPIO 15
Pin 3	NC
Pin 4	Gnd

<u>OLED</u>	<u>ESP32</u>
GND	GND
VDD	3.3V
SCK	SCK (GPIO 22)
SDA	SDA (GPIO21)

OLED AND DHT22 SENSOR

```
# Display the DHT22 Sensor Data on the 0.96" OLED Display
import machine, time, ssd1306
from machine import Pin, I2C
from ssd1306 import SSD1306_I2C
from time import sleep
import dht
sensor=dht.DHT22(Pin(15))
i2c=I2C(scl=Pin(22), sda=Pin(21))
oled=SSD1306_I2C(128,64,i2c,0x3c)
oled.fill(0)

while True:
    try:
        sensor.measure()
        print('Temp= %.2f' % sensor.temperature())
        print('Humidity = %.2f' % sensor.humidity())
        oled.text('Temp: %.2f' % sensor.temperature(),0,0)
        oled.text('Humid: %.2f' % sensor.humidity(),0,10)
        #oled.text("Hello World!",0,10)
        oled.show()
        sleep(3)
    except OSError as e:
        print('Failed to read sensor!')
```

b) Next, you can add in the Wifi connectivity:

c) Create a wifi_credential.py file to put your WIFI NETWORK SSID and password.

```
import network
import wifi_credentials
```

```
# Configure the ESP32 Wifi as STAtion
sta = network.WLAN(network.STA_IF)
if not sta.isconnected():
    print('connecting to network...')
    sta.active(True)
    sta.connect(wifi_credentials.ssid, wifi_credentials.password)
    while not sta.isconnected():
        pass
#print('network config:',sta_ifconfig())
```



```
wifi_credential.py
SSID='Your WiFi Network SSID'
Password='Your WiFi Network Password'
```

- d) Add in the Wifi connectivity to your script:

```
import urequests
# Setting Up Your ThingSpeak
HTTP_HEADERS = {'Content-Type':'application/json'}
THINGSPEAK_WRITE_API_KEY='0RHER0TEG6E1ALFX'
UPDATE_TIME_INTERVAL=5000 #ms
last_update=time.ticks_ms()

request=urequests.post('http://api.thingSpeak.com/update?api_key='+THINGSPEAK_WRITE_API_KEY,json=dht_readings, headers=HTTP_HEADERS)
request.close()
```

- e) Next, add in the codes to send data to Thingspeak using Restful API Post command. First, you will need to import the urequest library.
- f) Complete Thingspeak Post Request MicroPython Script but without the OLED display:

```
# ThingSpeak_Post.py
# ThingSpeak using RestAPI Post
# Import Libraries
import machine
import network
import wifi_credentials
import urequests
import dht
import time

# Define Your Connected Devices
led = machine.Pin(2, machine.Pin.OUT)
dht = dht.DHT22(machine.Pin(15))
```

```

# Configure the ESP32 Wifi as STAtion
sta = network.WLAN(network.STA_IF)
if not sta.isconnected():
    print('connecting to network...')
    sta.active(True)
    sta.connect(wifi_credentials.ssid,
wifi_credentials.password)
    while not sta.isconnected():
        pass
#print('network config:',sta_ifconfig())

# Setting Up Your ThingSpeak

HTTP_HEADERS = {'Content-Type':'application/json'}
THINGSPEAK_WRITE_API_KEY='0RHER0TEG6E1ALFX'
UPDATE_TIME_INTERVAL=5000 #ms
last_update=time.ticks_ms()

# Main Loop

while True:
    if time.ticks_ms() - last_update>=UPDATE_TIME_INTERVAL:
        dht.measure()
        t=dht.temperature()
        h=dht.humidity()

        dht_readings={'field1':t,'field2':h}

        request=urequests.post('http://api.thingspeak.com/update?api_key='+THINGSPEAK_WRITE_API_KEY,json=dht_readings,headers=HTTP_HEADERS)
        request.close()
        print(dht_readings)

        led.value(not led.value())
        last_update=time.ticks_ms()

```

5. Exercises

- i. Modify the code above to add in the OLED display to the ThingSpeak Rest API
- ii. Create a function `dht22( )` and put everything to read the DHT22 sensor inside the function `dht22( )`.
- iii. Can you observe the OLED display carefully. Did you spot any error in the display. Can you modify the OLED display portion of the code to solve the error, if any.

6. Reflection Questions

- i. Why must `sensor.measure()` be called before using `sensor.temperature()` and `sensor.humidity()`? What may happen if it is removed?
- ii. Why is `HTTP_HEADERS` set as `application/json` in the `urequests.post()` command? What may happen if the header is missing?
- iii. In the code below, how does ThingSpeak know which field is temperature and which field is humidity?
`dht_readings={'field1':t,'field2':h}`
- iv. What is the function of the `urequests.post()` command in the program?
- v. What is the purpose of the following code?
`request.close()`

7. Conclusion

In this laboratory exercise, we have:

-  Learn how to setup devices to be connected to ThingSpeak
-  How to display the sensor data in ThingSpeak using Rest API

- THE END -